

Analysis of overlapping origin–destination pairs between bus stations to enhance the efficiency of bus operations

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Jeongwook Seo¹, Shin-Hyung Cho², Dong-Kyu Kim¹ ✉, Peter Young-Jin Park³

¹Department of Civil and Environmental Engineering, Seoul National University, Seoul 08826, Republic of Korea

²Institute of Engineering Research, Seoul National University, Seoul 08826, Republic of Korea

³Department of Civil Engineering, Lassonde School of Engineering, York University, ON, M3J 1P3, Canada

✉ E-mail: dongkyukim@snu.ac.kr

Abstract: Public transit has a significant impact on minimising traffic congestion and reducing the cost of travelling in urban areas. It is necessary to evaluate the efficiency of the public transit operation in response to the individual traveller's demands for transit. This study aims to analyse the demand for transit with overlapping origin–destination (OD) pairs to enhance the efficiency of transit operations. To achieve this, disaggregated-level travel demand data, i.e. individual traveller's data are collected from an automatic fare collection system called smart card. The Kneedle algorithm is used to calculate the knee point of travel demand. The overlapping OD pairs, which are higher than the knee point value, are calculated and displayed in a map format. On the basis of the overlapping OD pairs, the demand-based overlap index for each bus route is defined to evaluate the efficiency of bus operations. The proposed method is applied to six districts with higher transit demands than other districts in Seoul. On the basis of the results, discussion on the action plans to enhance the efficiency of bus operations are presented. The method proposed in this study contributes to improving the efficiency of the bus system by reflecting individual users' travel demands.

1 Introduction

The rapid growth in urban sprawl gives public transit a vital role in relieving traffic congestion and reducing travel costs. The primary purpose of public transit is to provide quality transportation services to citizens so that people can travel conveniently with reasonable transit fares [1]. In 2017, Seoul's population was 9.8 million, with 924,200 veh/day average daily traffic volume across the city's 424 traffic analysis zones (TAZs). The city maintains 351 bus routes with 6244 bus stops. Among the various transportation modes (e.g. taxi, subway, passenger car, and bicycle), the bus has the most significant percentage (30.3%) of daily trips. Thus, the bus system must be as efficient as possible.

In the 1980s, the bus system was operated mainly to maximise the bus companies' profit [2, 3]. However, bus companies recently have begun to consider the users' perspectives when determining bus routes [4–8]. The most efficient bus routes should take care of both the operators' and the users' needs since transit efficiency is often measured as the sum of the company's operating costs and users' travel costs [4]. The term 'operating costs' refers to the cost of operating bus systems. Pattnaik *et al.* [9] stated that users' generalised travel costs consist of four components, i.e. (i) walking cost, (ii) waiting time, (iii) transfer cost, and (iv) in-vehicle cost.

Very few studies have applied advanced methods such as a genetic algorithm to search for the optimum set of bus routes that consider both the operators' and the users' needs [9, 10]. These studies posited that passengers' comfort and convenience also are crucial in determining the optimal operation of the bus transit system. In the past, many studies relied on data from surveys of passengers to understand the geographical and mobility characteristics related to a bus transit system. This approach was due to the difficulties associated with collecting the real data related to the use of buses such as individual travellers' origin–destination (OD) data [11]. Recently, due to the broad implementation of smart cards as an advanced automatic bus fare payment system in Seoul, more accurate and dynamic analyses of travel patterns can be conducted to evaluate a bus network system. Since smart cards allow researchers to directly access individual

travellers' OD data and other valuable sets of data (e.g. exact travel time), it is now possible to analyse the travel patterns [12–15].

Victor and Ponnuswamy [16] defined overlapping bus routes as bus routes that share route segments with the other routes. He suggested that the overlapping of bus routes should be minimised to provide a good quality transit service while considering the many operational constraints of transit companies, e.g. the number of operating buses per direction. Furthermore, the significant overlap of bus routes causes traffic congestion and high emission of greenhouse gases. We noted that the overlapping of bus routes, which is due to the overlapping travel demands for specific segments, is sometimes inevitable, especially for some areas, where the travel demand along the route is very high. However, most measures designed to evaluate the performance of overlapping bus routes only consider the physical lengths of the bus routes without proper consideration of a possibly even more important factor, i.e. travel demand. Since travel data can be collected from smart cards, it is now possible to consider real travel demand when analysing the overlapping of bus routes.

In this study, we developed a mathematical model using smart card data to assess overlapping bus routes. The smart card system provided the information on the number of trips between individual OD pairs, i.e. the real traffic demand, along specific bus routes. We used the Kneedle algorithm, which extracts a point needed for analysis after calculating the gradient between the maximum and minimum points. After that, we determined overlapping OD pairs through the data with demand above that point and developed the performance measures. The rest of this paper consists of four sections. The first section discusses the results of the literature review to provide an understanding of the previous approaches used to evaluate overlapping bus routes. The second section presents the specific design and methodology of our study. The third section provides a case study, in which we assessed the impact of overlapping bus routes. The fourth section presents our conclusions and recommendations for future research.

2 Literature review

This section focuses on reviewing studies that have attempted to evaluate the overlapping of bus routes while simultaneously considering travel demand.

An early study [2] used travel demand data (**OD** data) to allocate bus routes. The study had two main objectives: (i) to reduce overlapping travel routes and (ii) to operate the buses as close to maximum capacity as possible to maximise profits. The study did not consider service quality from the perspectives of users. An earlier study proposed a bus route allocation method for scheduling bus frequency as evenly as possible while reducing duplicated bus routes [17]. Diab *et al.* [18] analysed the impact of overlapping bus services on headways, delays, and bus bunching (bus bunching refers to tighter than scheduled headways between buses). They first developed a linear regression model that used headway delays to identify the influence of overlapping bus services. The headway delay (in seconds) was used as a dependent variable, and the number of overlapping trips was used as an independent variable. They also developed a binary logistic model to estimate the bunching of buses and reported that overlapping bus services were affected significantly by headway delays.

Some studies developed a kind of index to represent the extent of overlapping bus routes while evaluating the overall performance of the bus service [19, 20]. For instance, Katsampi *et al.* proposed a performance index for the bus system in Athens, Greece using the analytic hierarchy process in 2015. They considered three criteria, i.e. (i) efficiency and effectiveness, (ii) service, and (iii) design of the bus system. Efficiency and effectiveness were measured using passengers/shift, passengers/veh km, passengers/itinerary, cost/passenger, revenue/passenger, and accidents/1000 veh km. Service was determined by the number of complaints and the number of itineraries completed. The design was measured by directness, overlapping, and length of the bus routes. Among these indices, the overlapping route was calculated between line pairs as a percentage of their length, and the maximum value was used, which was measured as follows:

$$S^l(\%) = \max \left(\frac{\sum_k (d_k^l \times O_k^l)}{\sum_k d_k^l} \times 100 \right), \text{ all } l \neq j \quad (1)$$

where S^l is the overlapping route performance measure; d_k^l is the length of segment k of the bus route l ; and O_k^l is a dummy variable equal to one if a bus route is overlapped in segment k of the bus route j .

In Korea, overlapping bus routes in Seoul's bus system have been indexed and evaluated since 2004 [20]

$$R^l = \frac{\sum_k (d_k^l \times n_k^l)}{\sum_k d_k^l} \quad (2)$$

where R^l is the index of the overlapping bus route; d_k^l is the length of segment k of bus route l ; and n_k^l is the number of bus routes passing through segment k of bus route l . R^l refers to the average of d_k^l taking n_k^l as weight. Since segment k is at least the segment given by route l , n_k^l is 1 or more. The value of R^l is 1 or more. $R^l = 1$ when $n_k^l = 1$ for all k . If $R^l = 1$, the route has no redundancy, and the larger R^l is, the more overlapping the bus route becomes.

Equation (2) also has been used to evaluate the bus system of Daejeon and Incheon in South Korea [21, 22]. Gebeyehu and Takano [23] proposed a method to optimise the bus routes between urban and suburban areas using geographic information system to reduce overlapping bus routes. They tried to minimise both travel time and overlapping routes simultaneously. Dijkstra's algorithm was used to obtain decreased travel time for each route, and aggregated TAZ-based **OD** demand was used. To minimise overlapping routes, they considered three measures, i.e. (i) bus route area coverage, (ii) frequency of bus service, and (iii) capacity of the bus. The studies listed in this section focused only on the

operators' concerns, e.g. headway delay, bus bunching, and physical length of the route [16–18]. None of the previous studies evaluated overlapping bus routes considering individual travel demand to reflect individual users' perspectives. The studies analysed overlapping routes using aggregated travel demand (**OD** matrix between TAZs) or an even more coarsely aggregated approach based on regions (urban and suburban areas) [23]. None of the studies directly collected and reflected individual travellers' travel demand between **ODs**, i.e. none of the studies used a disaggregated (per person) level **OD** matrix.

In this study, we evaluated overlapping bus routes in Seoul, South Korea, using a new approach that considers individual travellers' real demand between origins and destinations. The objective of this study was to analyse and evaluate the degree of overlapping in bus routes from the perspective of user demand. Also, we proposed a segment-based overlapping route index (OL_S) and demand-based overlapping index (OL_D) and compared between them. The results of this study are expected to improve the efficiency of the Seoul bus system and improve convenience for both operators and users since the new method directly considers individual passengers' real demand and **OD** patterns.

3 Study design

3.1 Knee point analysis

The term p_{ij}^l denotes the number of passengers on bus route l , boarding at bus stop i , and alighting at bus stop j . Since this study analyses **OD** pairs using individual passenger data, too many **OD** pairs contain low p_{ij}^l values, e.g. **OD** pairs with p_{ij}^l less than or equal to 10 made up about 80% of the total data in this study. If all of the data were analysed without preprocessing, the results could be underestimated. It was, therefore, necessary to extract the data that contained more than a certain value of p_{ij}^l before continuing this study. That value of p_{ij}^l was the knee point which is defined as the point of maximum curvature in continuous data. The knee point can be estimated as

$$K_f(x) = \frac{f''(x)}{(1 + f'(x)^2)^{1.5}} \quad (3)$$

If the sign of $f'(x)$ is constant for all x , the knee point is the same as the inflection point. However, the inflection point does not represent the exact knee point when analysing discrete data. Satopaa *et al.* [24] developed the Kneedle algorithm, which can be used to determine the knee point in discrete data. The general procedure includes: (i) deciding the threshold value of the data, (ii) drawing a line segment that connects $(x_{\min}, y_{\max})$ and $(x_{\max}, y_{\min})$, i.e. the maximum and minimum y values of the data, and (iii) finding the knee point farthest from the line segment. Deciding the threshold value, which ends up being the same as the minimum value, is important to fit the line with increased accuracy.

X_{ij}^l denotes a unique identifier (ID), containing bus route l , boarding at bus stop i , alighting at bus stop j , and p_{ij}^l . $N_p(x)$ denotes the number of X_{ij}^l , whose p_{ij}^l value is equal to or greater than x . The Kneedle algorithm finds the knee point of p_{ij}^l . The procedure for finding the knee point in this study was as follows: (i) sort the X_{ij}^l in descending order of p_{ij}^l ; (ii) remove any X_{ij}^l that contains less than three passengers/h; and (iii) determine the knee point by using the Kneedle algorithm. By sorting X_{ij}^l in descending order of p_{ij}^l , graphs are obtained for each AM or PM peak hour and district, as shown in Fig. 1. A is the maximum point of y , B is the threshold point, K is the knee point, and c is the knee point value of p_{ij}^l . In all cases (both district and time period), the number of X_{ij}^l with p_{ij}^l values of 1 or 2 exceeded 50%. In other words, X_{ij}^l with a low p_{ij}^l value could underestimate the value of the knee point. Thus, we decided to use the threshold value as three passengers/h.

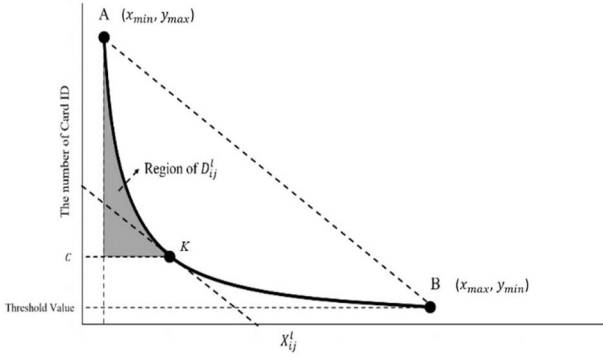


Fig. 1 Kneedle algorithm

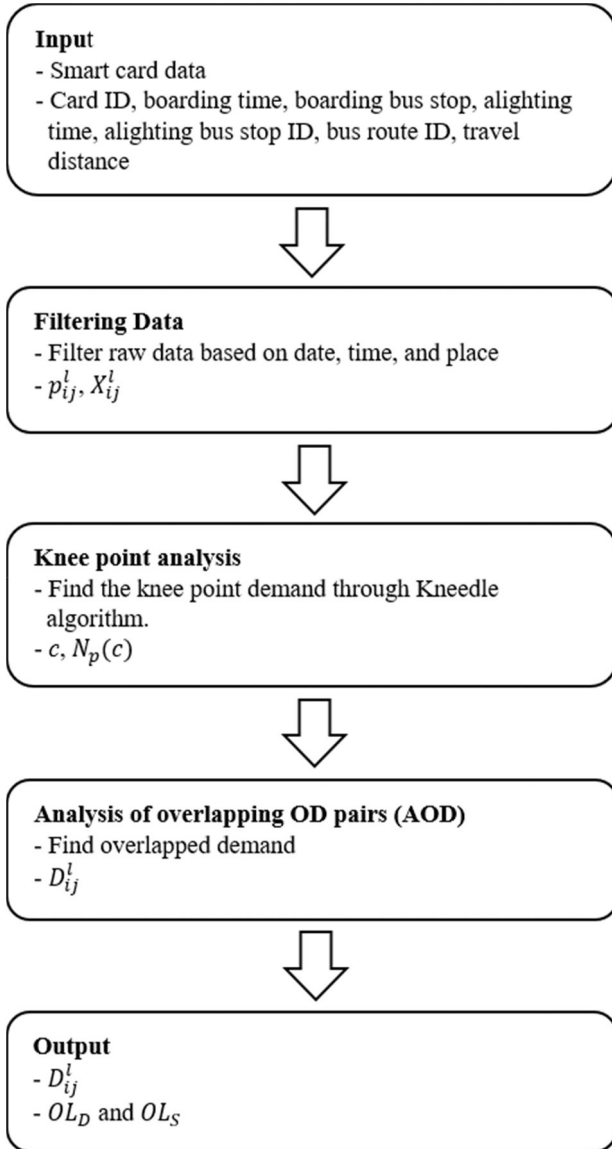


Fig. 2 Procedure of analysing overlapping OD pairs

3.2 Overlapping OD pairs

After determining the knee point of p_{ij}^l , the demand for overlapping OD pairs is obtained using X_{ij}^l . The procedure was: (i) choose the X_{ij}^l with p_{ij}^l greater than or equal to c ; (ii) determine whether there is more than one X_{ij}^l with the same origin and destination, but a different bus route; and (iii) remove all values of X_{ij}^l that do not have more than one different bus route. Then, we define the p_{ij}^l of the remaining X_{ij}^l as D_{ij}^l , which is the number of overlapping

demands on bus route l , boarding at bus stop i , and alighting at the bus stop j . D_{ij}^l is calculated using (4)

$$D_{ij}^l = p_{ij}^l \quad (4)$$

s.t.

$$\begin{aligned} p_{ij}^l &\geq c \quad \forall i, j, l \\ l_{ij}^a &= l_{ij}^b \quad \text{for } a \neq b \end{aligned}$$

where l_{ij}^a is the bus route, l^a indicates passengers passing through from bus stop i to bus stop j , and D_{ij}^l is equal to p_{ij}^l when there are overlapping bus routes that have a value equal to or greater than c

$$D_{ij} = \sum_l D_{ij}^l \quad (5)$$

The result of adding all D_{ij}^l for l , D_{ij} is total overlapping demand boarding at bus stop i , alighting at bus stop j , i.e. D_{ij} is the number of trips of OD pair, where its origin and destination are i and j , respectively. We call the OD pairs of D_{ij} overlapping OD pairs. Fig. 2 presents the overall process of calculating the overlapping OD pairs.

We analysed the overlapping OD pairs, conducted an analysis of overlapping OD (AOD) pairs by each type of bus, and compared the demand-based overlapping index (OL_D) and the segment-based overlapping index (OL_S). The descriptive statistics of overlapping OD pairs was developed such as the average length between origin and destination, and these statistics described the overall characteristics of each district. Besides, three districts were visualised on maps. Since each bus type has a different role in transporting passengers, we applied the methodology of AOD according to the type of bus. Arterial, feeder, and local buses were analysed to evaluate the roles of each type of bus. Arterial buses operate across districts, feeder buses operate within one district, and local buses operate between neglected areas and subway stations or bus stops. The knee point and descriptive statistics of the overlapping OD pairs were determined for each type of bus. On the basis of these results, it is discussed whether the role by the type of bus was performed well.

As discussed, we defined OL_S^l and OL_D^l to evaluate the efficiency of the bus route based on the segment-based overlap of the bus route and the demand-based overlap of the bus route. We defined $(2)R^l$ as the segment-based overlapping index of bus route l (OL_S^l), i.e.

$$OL_S^l = \frac{\sum_k (d_k^l \times n_k^l)}{\sum_k d_k^l} \quad (6)$$

where d_k^l is the length of segment k of bus route l and n_k^l is the number of bus routes that passes through segment k of bus route l .

There are only two indices to measure bus route overlapping. These are shown in (1) and (2). Equation (1) only explains whether there are segments with overlapping bus routes. Equation (2) is used to determine how many bus routes overlap on the segments. As we discussed in this paper, the Daejeon Research Institute used the 2004 overlap index in 2008, and the Incheon Research Institute used this index in 2016. For that reason, we used the 2004 index as our segment-based overlap index.

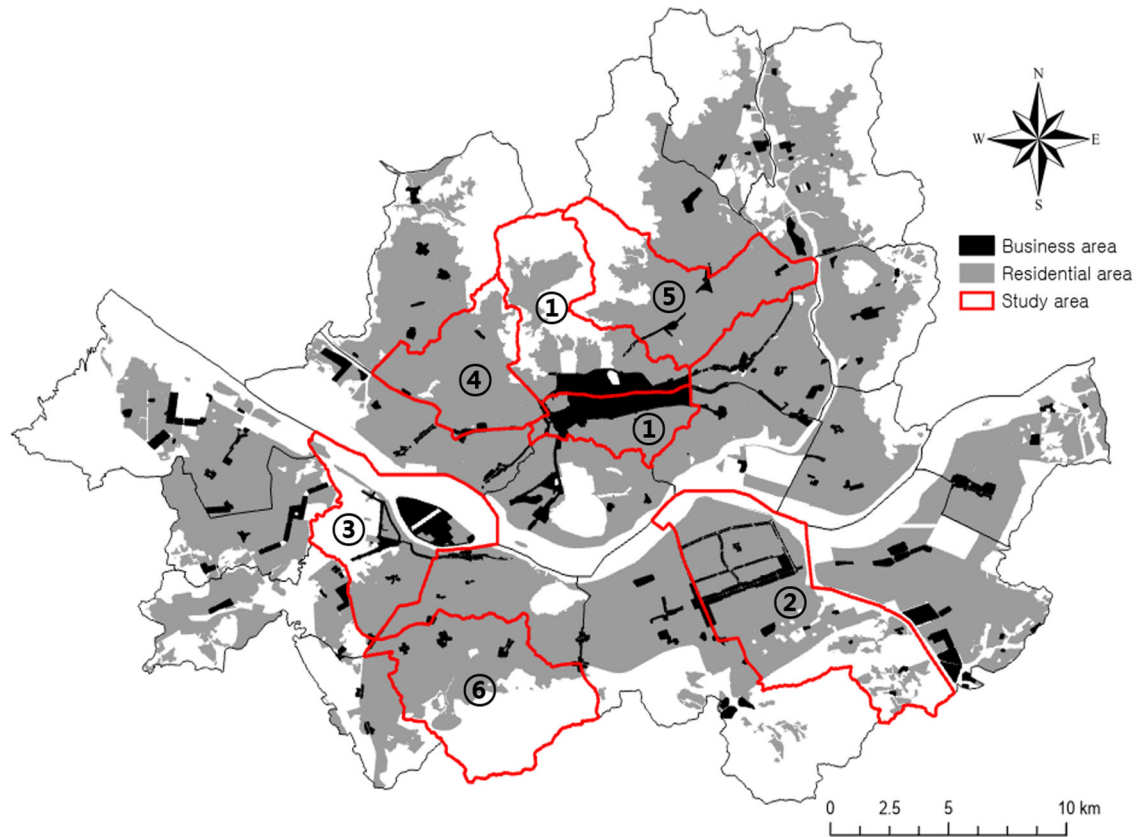
OL_D^l , the demand-based overlapping index of bus route l is calculated as follows:

$$OL_D^l = \frac{\sum_{(ij)} D_{ij}^l}{\sum_{(ij)} p_{ij}^l} \times 100 (\%) \quad (7)$$

where OL_D^l is the overlapping demand of bus route l as a percentage of the total demand for bus route l at peak hours. That

Table 1 Contents of smart card data

Information	Description	Data type
card ID	card number for each smart card	object
boarding time	boarding time including date	int (ten digit)
boarding bus stop ID	given number of boarding bus stop ID	int (ten digit)
alighting time	alighting time including date	int (ten digit)
alighting bus stop ID	given number of alighting bus stop ID	int (ten digit)
bus route ID	given number of every bus route	int (ten digit)
travel distance	distance from the boarding bus stop to alighting bus stop	int (ten digit)

**Fig. 3** Study area in Seoul

is, OL_D^l is the ratio of the sum of D_{ij}^l for all i and j to the sum of p_{ij}^l for all i and j at peak hours. Graphs with OL_S^l as the x -axis and OL_D^l as the y -axis were analysed by region and type of bus. For both indexes, the higher the value, the greater the degree of overlap.

4 Case study

4.1 Data description

The smart card, i.e. the automatic fare payment system in Seoul, collected about 94% of the bus fares paid by users in 2016. In this study, we used individual passengers' smart card data collected on 17 May 2017. The data contain more than 20 different sets of information, but, for this study, only seven sets of information were used, as listed in Table 1. The values of p_{ij}^l were calculated by adding the number of card IDs that had the same values as the boarding bus stop ID, alighting bus stop ID, and bus routes ID. The boarding time and the alighting time were used to extract peak-hour data, and the boarding bus stop ID and alighting bus stop ID were used to extract the selected study area. We defined the AM peak hours as 7:00 to 10:00 AM and the PM peak hours were from 5:00 to 8:00 PM. Therefore, the non-peak hours were 11:00 AM until 3:00 PM.

Fig. 3 shows land use for Seoul in 2010. Seoul consists of 26 districts, and the black areas, grey areas, and white areas represent

business areas, residential areas, and other areas, respectively. The red boundary shows the study areas. The case study examined six of 26 districts in Seoul, i.e. Jongno-Jung (district 1), Gangnam (district 2), Yeongdeungpo (district 3), Seodaemun (district 4), Seongbuk (district 5), and Gwanak (district 6). The first three districts are the central business districts (CBDs). The other three districts have the highest traffic demand among the 22 districts, except for the CBDs. Traffic congestion occurs regularly in these six districts, particularly during peak hours, emphasising the importance of an efficient transit system.

Individual passengers' travel demands/h (to or from a bus stop in each district) for each district for all day, morning peak hours, afternoon peak hours, and non-peak hours are shown in Table 2. The unit for travel demand is passengers/h. District 1 had the highest all-day travel demand, and there were no significant differences in demand between peak and non-peak hours. Other districts had higher demand during the two peak-hour periods than during the non-peak hours. District 3 had the most significant difference in demand between the peak and non-peak hours. The other four districts had more demand during the AM peak hours than the PM peak hours. Districts 2 and 3, two of the three CBD districts, had higher travel demand during the peak hours than the non-CBD districts.

Table 2 Travel demand by district and time period

District		1 Jongno·Jung	2 Gangnam	3 Yeongdeungpo	4 Seodaemun	5 Seongbuk	6 Gwanak
travel demand, (passengers/h)	all day	27,719	21,422	15,605	17,951	17,521	18,157
	AM peak hour	28,636	37,014	40,617	33,645	28,806	33,645
	PM peak hour	26,063	35,312	42,082	29,035	27,532	28,246
	non-peak hour	28,240	19,145	13,461	16,386	16,566	15,594

Table 3 Knee points of $p_{i,j}^l$ and $N_p(c)$ at peak hour

District	Peak hours	Knee point of $p_{i,j}^l(c)$	$N_p(c)$
1	AM	32	254
	PM	50	72
2	AM	55	190
	PM	39	178
3	AM	56	136
	PM	34	184
4	AM	56	91
	PM	39	137
5	AM	47	131
	PM	45	102
6	AM	48	227
	PM	39	180

4.2 Calculation of knee point

We applied the Kneedle algorithm to each AM and PM peak hours at each district. Since the threshold value is three passengers/h, and each peak hour is 3 h, $p_{i,j}^l = 9$, nine is the threshold value in this study. The numbers of $X_{ij}^l(N_p(c))$ that are equal to or higher than the knee point of $p_{i,j}^l$ are shown in Table 3. The X_{ij}^l , which contains $p_{i,j}^l$ values equal to or greater than c for all CBDs includes destination as the district at the AM peak hour and origins as its district at the PM peak hour. The existence of extreme high $p_{i,j}^l$ values compared with others causing a high value of c in each case. On the contrary, too many low $p_{i,j}^l$ values bring out a low value of c in each case. All districts except district 1 have higher c value at the AM peak hour than PM peak hour. Both the values of c and $N_p(c)$ are higher at AM peak hour than PM peak hour in districts 2, 5, and 6. It indicates that many X_{ij}^l contain high values of $p_{i,j}^l$. The values of c are higher at AM peak hour than PM peak hour, but the values of $N_p(c)$ are lower at AM peak hour than PM peak hour. In other words, few X_{ij}^l contain high values of $p_{i,j}^l$. It implies that many passengers travel to similar areas. The value of c is larger during the PM peak hours, but the value of $N_p(c)$ is larger during the AM peak hours in district 1. Too many low $p_{i,j}^l$ values result in a low value of c in AM peak hour. Furthermore, district 1, which is located at the centre of Seoul, is one of the CBDs, and the transportation hub of Seoul. Considering that passengers are moving toward the CBDs in the morning, this indicates that passengers move from diverse areas to district 1 in the morning.

4.3 AOD pairs

In Seoul, the bus travel pattern in the CBD areas shows high demand between subway stations in the CBD areas and core business areas or schools during the AM peak hours in particular. The large portion of the travel demands in the CBD areas occurs from core business areas or commercial areas to subway stations during the PM peak hours. Furthermore, large travel demands from residential areas to subway stations or schools take the largest portion of bus trips during the AM peak hours in non-CBD areas. The majority of bus travel demands during the PM peak hours in non-CBD areas are from subway stations to residential areas. Also, the origins and destinations of most overlapping OD pairs are in one district.

Table 4 shows the number of overlapping OD pairs and the descriptive statistics of D_{ij} . Since it is located in the centre of Seoul, passengers take various bus routes, which lead to more overlapping OD pairs having lower D_{ij} values than other CBD areas at the AM peak hours, and only one overlapping OD pairs at the PM peak hours in district 1. In other words, passengers move in various directions. For the same reason, the average length of the overlapping OD pairs in district 1 was longer than those of the other two CBDs.

Although the average distance of bus travel during the AM peak is shorter than it is during the PM peak, the AM peak has a higher number of overlapping OD pairs and a higher D_{ij} value than the PM peak hours. It means that there is a higher number of passengers during the AM peak, but they travel a shorter distance than in the PM peak. In district 3, the number of overlapping OD pairs is similar for the AM and PM peak hours. This district also had a shorter average length and a higher average value of D_{ij} during the AM peak hours. In district 4, even though the number of overlapping OD pairs is similar to district 3, the average and standard deviation of D_{ij} are higher. In district 3 and district 4, the origins and destinations of overlapping OD pairs in the AM peak hours correspond almost to the destinations and origins of overlapping OD pairs in the PM peak hours. It refers to the fact that the passengers' patterns are relatively constant compared with other districts. Even though district 5 has a lower number of overlapping OD pairs and lower D_{ij} values during the AM peak hours than the PM peak hours, these values are smaller than the values for the other five districts. The number of overlapping OD pairs and the values of D_{ij} were the highest in district 6. It refers to the fact that the overlapping travel demand is the most severe.

Two or more overlapping OD pairs, which are overlapped consecutively, or an overlapping OD pair having much higher D_{ij} value than the other pair(s) are selected as a segment. These segments with one or more overlapping OD pairs of three districts (districts 1, 2, and 6) are shown in Fig. 4. The AM peak results are presented on the left-hand side, and the PM peak results are displayed on the right-hand side. On the resulting maps, the lowest D_{ij} values are shown by white lines. As the values of D_{ij} increase, the line becomes darker. District 1 was shown on the map because the number of overlapping OD pairs in the AM peak hours was very different from the PM peak hours. District 2 was shown on the map and district 3 was not shown because district 3 is located near the river, which leads to constraints on the traffic pattern. However, district 2 was observed to have better results because it is composed of a grid road network, and there are no geographical constraints around it. District 6 was selected because it had the largest number of overlapping OD pairs. Districts 4 and 5 are excluded, because for several reasons. These two districts are not CBD, do not have many overlapping OD pairs compared with districts 2 and 6, and do not have considerable differences between AM peak hour and PM peak hour such as district 1. Table 5 provides the sum of D_{ij} and the number of overlapping OD pairs for each segment shown in Fig. 4. Fig. 4a shows that the overlapping OD pairs were dispersed when there was low demand in the AM peak hours. For example, segments 1 and 2 have high D_{ij} values, i.e. 684 and 490, respectively. Segment 1 consists of three overlapping OD pairs and segment 2 consists of two overlapping OD pairs. In the PM peak hours, there was only one overlapping OD pair. It contained 242 demands, and it was located in the same place as segment 1.

Fig. 4b shows that there were more overlapping OD pairs with high D_{ij} values during the AM peak hours than there were during

Table 4 Descriptive statistics of overlapping OD pairs and D_{ij}

District	CBD								Non-CBD			
	1		2		3		4		5		6	
Peak hours	AM	PM	AM	PM	AM	PM	AM	PM	AM	PM	AM	PM
number of overlapping OD pairs	33	1	24	16	11	13	12	15	5	7	47	42
D_{ij} mean	141	242	202	157	195	148	279	161	121	149	309	222
standard deviation	93	0	85	79	52	77	188	98	24	61	254	167
average length, km	1.82	2.11	0.91	1.30	1.48	1.21	1.50	2.23	2.48	2.18	2.18	2.07

the PM peak hours. Segments 3 and 5 have remarkably high D_{ij} values, i.e. 573 and 615, respectively. The value of D_{ij} on segment 4 was 1276 in the AM peak hours. Even though this segment only consisted of one overlapping OD pair, it had high D_{ij} values because this street passes the core business district in district 2. In the PM peak hours, though segment 6 consisted of only one overlapping OD pair, it showed 427 travel demands, about 2.4 times higher than the average D_{ij} value.

Fig. 4c shows the results for district 6. Even though segments 8–13 are in district 6, which is a non-CBD area, the demand for bus system was at a significant level. Especially, segment 10 had the most passenger use of the bus transit system due to the lack of subway line and traffic congestion. The segments pass mainly through residential areas located in a mountainous area. Owing to the high altitude of the residential area, it is difficult to reach without taking a bus or car from/to the nearest subway station (noted as α in Fig. 4c). Therefore, many overlapping OD pairs have high D_{ij} values from residential areas to subway stations in both the AM and PM peak hours. Segments 9 and 10 are from residential areas to the subway stations. Segment 8 contains seven different bus stops in this residential area and ends at another subway station at the boundary of district 6. During the PM peak hours, the nearest subway station to the residential area and residential area to the nearest subway station belong to segments 11 and 12, respectively. Segment 13 is from a university to a subway station. It is the closest subway station to the university.

5 Discussion

5.1 Comparison of AOD

The AOD methodology was conducted in district 2 for three different types of buses: arterial, feeder, and inter-regional. Table 6 presents the results. In district 2, there were 39 arterial bus routes, 29 feeder bus routes, and 20 inter-regional bus routes.

The number of overlapping OD pairs for arterial buses is more than double the number for feeder buses. Although arterial buses use main roads and feeder buses, use roads with low traffic volume and few lanes, the average length of the arterial and feeder bus routes was similar. The D_{ij} value for arterial buses ranged from 77 to 1276 and the D_{ij} value for feeder buses ranged from 62 to 269. These results indicate the role of each type of bus. Since district 2 is a CBD, more passengers take arterial buses than feeder buses. On the other hand, since both the origin and destination of the overlapping OD pairs for arterial buses are located in district 2, it is clear that some arterial buses perform the role of a feeder bus. Very few inter-regional buses carry more than nine passengers during the peak hours, and as inter-regional buses are organised to avoid overlapping routes, no inter-regional buses have overlapping OD pairs during either of the peak hours.

5.2 Demand-based overlapping index (OL_D)

We evaluated whether OL_D is related to OL_S . OL_S values were calculated only for bus routes, where OL_D values exist. In other words, the OL_S values were calculated only if the bus routes contained D_{ij} . If there was a value of OL_D in both the AM and PM peak hours, we calculated OL_D by adding the data from both peak hours. A larger value for a measure indicated a greater degree of overlapping. Fig. 5 shows the relationship between the two indices in both peak hours. Only bus routes that included overlapping OD

pairs were used to calculate OL_D . Owing to the different shapes of graphs on each bus type and district, we analysed Fig. 5 according to the type of bus. Local buses usually had up to four OL_S values with varying OL_D values. The role of a local bus is to provide transportation service to citizens living in underserved areas, where it is difficult to operate feeder or arterial buses, usually because of geographical and/or financial constraints. A local bus can deliver passengers from underserved areas to nearby subway stations or bus stops. Owing to its unique role, it is more important for a local bus to consider passenger demand than overlapping routes using OL_S . As OL_D has a wide range of values, the quality of a local bus can be improved through further evaluation for bus routes that have high OL_D values.

The overall OL_D values in district 1 are lower than those in districts 2 and 6. As district 1 is not just a CBD in Seoul, but also the main transportation hub of the city, more arterial buses pass through this district than any other district in the city. For that reason, in district 1, the OL_S values of the arterial buses are widely spread compared with the OL_D values of arterial buses. However, the OL_D values of the feeder buses are widely spread compared with the OL_S values of the feeder buses (because the role of feeder buses is to transport passengers within a particular district). In other words, the graphs of arterial buses and feeder buses are different: arterial buses tend to form lines parallel to the x-axis and feeder buses tend to form lines parallel to the y-axis.

In district 2, bus routes have OL_S values of less than six, except for six of the 47 arterial bus routes. The values of OL_D range from 0.8 to 22 and the mean is 5.8. The six arterial bus routes have OL_S values greater than 6, but low OL_D values in district 2. This indicates that the demand for the six arterial bus routes is much higher than the demand for the other bus routes even though the six routes do not physically overlap each other.

In district 6, most OL_D values of feeder bus routes are between 5 and 10, but about 50% of feeder bus routes have more than 12 OL_S values, which value is rare in the other two districts. This means that even though the feeder bus routes are well-located with the adequate overlapping segment, demands for those bus routes are very high and there maybe a problem with providing enough capacity to meet demand. These bus routes should be evaluated by examining the bus dispatch times to assess the efficiency of the bus operating system. The OL_D and OL_S patterns for arterial and feeder buses are similar, indicating that these two types of service essentially play the same role rather than different roles in district 2. In district 6, both arterial buses and feeder buses have relatively high OL_S values (between 5 and 10). This is largely due to the geographical characteristics of district 6, as mentioned earlier. More than half of the feeder bus routes in district 6 have OL_D values of 12 or more, whereas the OL_D values for all arterial bus routes in district 6 are less than 8. The feeder buses transport passengers from the residential areas to the subway station, while the arterial buses transport passengers from easily accessible areas such as subway stations, to other districts.

5.3 Suggested action plan

Fig. 6 shows the OL_D and OL_S of bus routes in all six districts. The bus routes were divided into four groups according to the average value of OL_D (5.56) and the average value of OL_S (5.80).

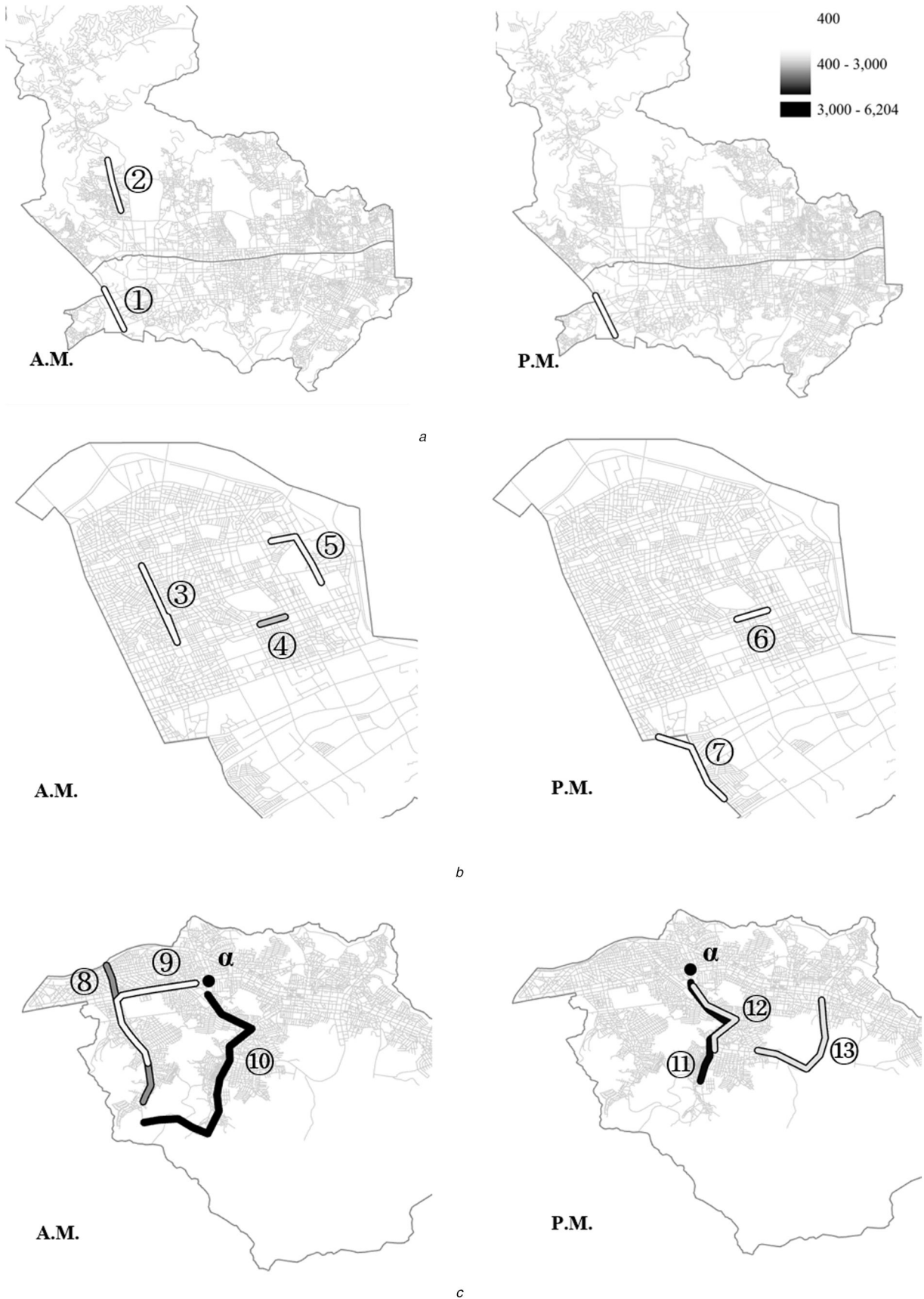


Fig. 4 Results of analysis of AOD for the AM peak (left) and the PM peak (right)
(a) District 1, (b) District 2, (c) District 6

For group 1 bus routes, the OL_D is higher than the average OL_D , but OL_S is lower than the average OL_S . This pattern indicates that even though the length of the overlapping routes' segment is short, many passengers use that segment. Passengers face inconvenience at bus stops due to the increased dwell time and waiting time, and

they experience the congestions on buses. Increasing the capacity of the bus stops and reducing time headway of buses is required to ensure a more efficient and convenient bus route operation.

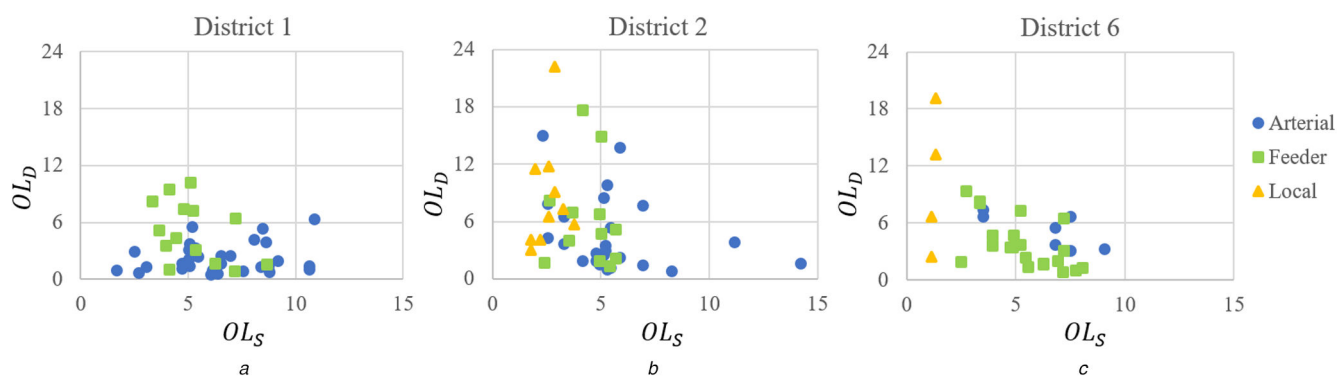
For group 2 bus routes, OL_D and OL_S are both higher than the average OL_D and OL_S . This pattern could indicate traffic

Table 5 Demands of segments of overlapping OD pairs

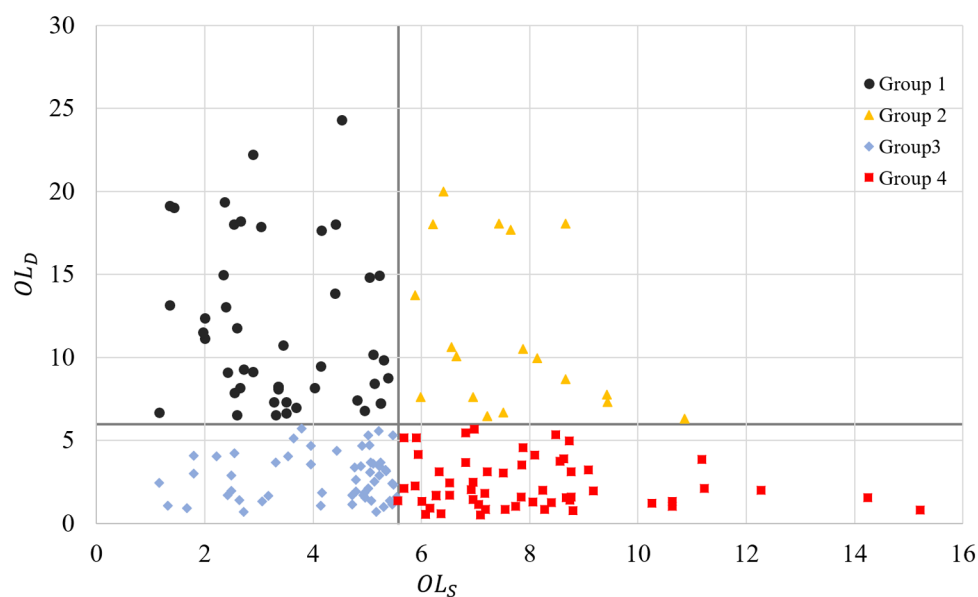
Segment	Sum of D_{ij} , passengers	Number of overlapping OD pairs
1	684	3
2	490	2
3	573	2
4	1276	1
5	615	3
6	427	1
7	590	2
8	1752	7
9	542	3
10	6204	10
11	2917	6
12	866	3
13	889	3

Table 6 AOD methodology of each type of bus in district 2

Type of buses	Arterial		Feeder		Inter-regional	
Peak hours	AM	PM	AM	PM	AM	PM
knee point of $p_{i,j}^l(c)$	34	22	30	29	16	9
number of overlapping OD pairs	35	40	14	10	0	0
D_{ij}	average		average		average	
	standard deviation		standard deviation		standard deviation	
	average length, m		average length, m		average length, m	
	187	101	138	100	0	0
	206	54	57	32	0	0
	910	1165	921	1372	0	0

**Fig. 5** Comparison of OL_S and OL_D for types of buses

(a) District 1, (b) District 2, (c) District 6

**Fig. 6** OL_S and OL_D divided into four groups

congestion and bus delays due to the high degree of overlapping routes, and could also indicate inconvenience for passengers at bus stops and on buses for that same reason as group 1. Implementation of exclusive bus lanes such as high-occupancy vehicle (HOV) lanes can reduce both operator and user costs. Also, operational efficiency maybe improved if a separate circulation bus route targeting overlapping segments is introduced, and the time headway is increased on existing overlapping bus routes.

For group 3 bus routes, OL_D and OL_S are lower than the average OL_D and OL_S . This pattern could indicate that bus routes in this group are well-dispersed, covering a broad area and transporting passengers from various origins to various destinations. Therefore, the bus routes of this group are operating efficiently.

For group 4 bus routes, OL_D is below the average, but OL_S is over the average. This pattern could indicate inefficient operations as it implies low demand and long, overlapping routes. This situation could lead to bus stops that are over capacity, and could also cause traffic congestion resulting in increased operating costs for bus companies. The problems can be reduced by reorganising the bus routes or by assigning group 4 bus routes to other routes and reducing the time headway.

6 Conclusion

Many previous studies have tried to optimise bus routes considering passengers' perspectives, but no studies have evaluated the overlapping of bus routes based on the passengers' perspective. This study aimed to analyse overlapping bus routes to reflect the individual traveller's demand. Six districts including three CBD areas in Seoul, South Korea, were selected as the study area. Both AM and PM peak hours were considered. We used the Kneedle algorithm to calculate the knee point, extracted the $p_{i,j}^k$ above this knee point, and calculated D_{ij} . D_{ij} was analysed statistically and plotted on a map. Three types of buses were analysed using the AOD method. The arterial bus had a greater number of overlapping OD pairs than the feeder bus. The roles of the three types of buses, i.e. arterial, feeder, and local buses, were evaluated based on OL_D and OL_S values. Owing to the role of local buses, OL_D is a more effective index to evaluate local bus than OL_S . The feeder buses and arterial buses in district 1 showed different tendencies. Most buses had concentrated OL_S values, but, in district 2, they had various OL_D values. Feeder buses had high OL_D values. On the basis of the OL_D and OL_S values, we proposed a way to increase the efficiency of the bus routes after dividing the bus routes into four groups. Therefore, comparing OL_D and OL_S helps to evaluate the efficiency of bus routes.

Although this study has the limitation that the AOD method alone cannot provide a complete evaluation of a bus system, the results are useful and show that the AOD method reflects passengers' perspectives about overlapping bus routes. Furthermore, OL_D helps evaluate the performance of each type of buses. Future research applying this methodology in the entire Seoul area and/or in other cities could be useful and effective to operate the bus system. Besides, studying the available algorithms for the optimisation of bus routes including the demand-based overlap of bus routes, could contribute to the efficiency and comfort of bus systems.

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